

# Argmax Decoding Causes Liquidity Failure in a Market of Language-Model Traders

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## Abstract

We populate a continuous double auction with traders whose decisions are made by a typed language model that returns a probability distribution over actions, and vary only how that distribution is consumed. Taking the model’s modal action—the default in virtually every deployment—makes the market stop trading after shocks: following a positive shock, the market clears in 65.6% of periods against 91.3% after a negative shock, a gap of 25.8 percentage points (95% CI [17.3, 34.2],  $t = 6.37$ ,  $n = 20$  preregistered). Sampling from the same distribution reduces the gap to 3.1 points. Two algorithmic baselines that are symmetric by construction show no gap. The mechanism is direct and observable: 100% of non-trading periods under modal decoding had one side of the order book entirely empty, against 3.4% for the zero-intelligence baseline. Because every trader conditions on the same public signal, a deterministic policy makes them act identically, and a market of unanimous buyers has no counterparty. The effect decomposes into two parts: unanimity sets its magnitude, while a standing directional bias in the arm’s order flow determines which shock it strikes.

We test this mechanism with a preregistered prediction it could have failed: if unanimity is the cause, the effect must vanish as the market grows. It does, from 37.0 points at four traders to 4.2 at thirty-two ( $-0.112$  per doubling,  $t = -6.91$ ). A companion prediction—that sampling would show no such decay—was falsified: sampling herds too, seven times more weakly, on the same curve. We therefore report a single mechanism whose strength is set by how sharply the decode rule collapses the model’s distribution, rather than two regimes.

Separately, we find that ordinary domain wording induces a 25-point asymmetry in the model’s stated conviction between buying and selling at equal mispricing, which vanishes under mirror-image phrasing. A preregistered test found this does *not* move pricing error, and we report that null.

All experiments were preregistered, including the two hypotheses that failed.

## 1 Introduction

A language model asked to choose among actions returns, in principle, a distribution. Almost every deployment discards it and takes the mode. For a single decision this is correct: the mode is the most likely action. This paper asks what happens when many agents driven by the same model make related decisions at the same time.

The answer, in a market, is that the market stops. Modal decoding makes agents conditioning on shared information act identically. In a setting that requires counterparties, unanimity is not consensus; it is the absence of trade. We measure this in a continuous double auction, isolate the mechanism, and confirm it with a prediction that could have failed and did not.

Our contributions:

1. **A liquidity failure caused by decoding, not by reasoning.** The model’s modal action is *correct* on both sides at every mispricing we tested. Nothing about its judgement is wrong. The failure is entirely in the aggregation of correct decisions.
2. **A directly observed mechanism, in two parts.** We instrument both sides of the order book and show that under modal decoding, non-trading periods lack a counterparty essentially always, while for symmetric baselines they do not. Unanimity sets the magnitude; a standing directional bias in order flow sets which shock it strikes. Our own first account used only the former and is corrected in Section 6.
3. **A falsifiable size-scaling test.** The mechanism implies the effect must decay with market size at a rate set by the agreement rate. We registered this before running it; it held.
4. **A prompt-sensitivity result, and a null.** Domain-idiomatic wording induces a large conviction asymmetry invisible to anyone reading only the modal action. A preregistered test found it does not move pricing error.

We emphasise the failures. Three preregistered predictions in this project were wrong, and each is reported: the original mechanism (Section 5), the sampling-amplification prediction, and the claim that sampling would show no size decay (Section 7). The result that survived did so against tests designed to kill it.

## 2 Related work

**Market design and baselines.** The experimental market tradition begins with Smith [1962]. Our lower baseline is the zero-intelligence trader of Gode and Sunder [1993], whose result—that budget-constrained random traders achieve high allocative efficiency—makes a random arm a genuine floor rather than a strawman, and motivates the value clamp that prevents any arm in our design from knowingly trading at a loss.

**LLM agents in simulated markets.** Lopez-Lira [2025] builds a simulated stock market with a persistent order book in which heterogeneous LLM agents submit structured decisions, and reports that agents adhere consistently to assigned strategies while the market reproduces price discovery, bubbles and underreaction. That work also notes that prompts can generate correlated behaviours affecting market stability, which is the phenomenon we isolate and measure here.

**Correlation as systemic risk.** Closest to our result is Ross et al. [2026], who argue that shared training and architecture make more capable models behave more similarly, creating a non-diversifiable risk floor, and who test this with LLM traders of varying capability in an agent-based market. They find correlated behaviour that *increases* with capability, and that it becomes a liability when agents share a misinformation environment.

Our contribution is complementary and, we think, orthogonal in a useful way. They vary the model and attribute correlation to properties of the model—its training, its architecture, its capability. We hold the model *fixed* and vary only how its output distribution is consumed. The correlation we measure is therefore not a property of the model at all but of a deployment choice,

and it is one a practitioner controls directly: our argmax and sampling arms issue identical requests and read identical responses, differing only in a single line of client code. Where Ross et al. [2026] identify a risk floor set by model similarity, we show a large component of realised correlation sitting above that floor, removable without changing models.

**Herding and cascades.** The classical account of correlated action is the informational cascade of Bikhchandani et al. [1992], in which an agent rationally ignores private information after observing enough predecessors. Our mechanism is deliberately simpler and should not be confused with it. Agents in our market do not observe one another’s decisions and draw no inference from them; they condition on a common public signal and a deterministic policy maps that signal to the same action. There is no learning, no sequence, and no inference—only a degenerate policy. The resulting unanimity looks like herding and has the same market consequence, but it arises without any of the cascade machinery, which is why it can be switched off at the decoder.

**Decoding and diversity.** That greedy decoding collapses output diversity is long established in text generation: Holtzman et al. [2020] show that maximisation-based decoding yields repetitive, degenerate text and introduce nucleus sampling as a remedy. The quality–diversity tension is well understood *within* a single generation. We report a consequence one level up: when many agents share a decoder, the diversity that sampling preserves is not a stylistic property of one output but the raw material a market needs to clear. Sampling is individually worse and collectively necessary.

**Confidence and calibration.** Kadavath et al. [2022] study whether language models can predict the correctness of their own answers, and subsequent work has documented systematic overconfidence in verbalised confidence. Our Experiment 5 touches this literature from an unusual angle: the model we use returns a probability distribution natively rather than a verbalised estimate, and we find that ordinary domain wording shifts that distribution by 25 points without changing the selected option. Any calibration measurement taken over such a model inherits whatever asymmetry the question’s phrasing introduced.

**What we could not find.** We did not find prior work isolating the decode rule as a cause of market-level liquidity failure, nor a preregistered size-scaling test of a herding mechanism in an LLM-populated market. We would welcome pointers; this survey was conducted by the authors and is not exhaustive.

## 3 The instrument

### 3.1 Market

A continuous double auction in one abstract good. Prices are integer ticks; matching is price-then-time priority; an incoming order executes at the resting order’s price. A trader’s resting orders are withdrawn at the top of its turn, so each trader has at most one live order. Submitting on one side cancels the trader’s resting orders on the other, preventing wash trades without leaving a crossed book.

Accounts are conserved by construction. Each trade moves cash from buyer to seller and inventory from seller to buyer, one for one. Short sales and borrowing are disallowed, and every

order is pre-checked against uncommitted cash or uncommitted inventory. A property test submits 20,000 random orders and asserts after every one that total cash and total units are exactly their endowed values, that no account is negative, that no trader has committed more than it holds, and that the book is never crossed.

Order size is fixed at one unit, so aggressiveness is the only lever an arm has; position sizing is a second channel and is deliberately disabled.

## 3.2 Fundamental and information

The fundamental  $F_t$  is piecewise constant. For the size and liquidity experiments we use a *matched-jump* process: jumps are evenly spaced, of identical magnitude, strictly alternating in direction, and even in number, so that every up window is paired with a down window of the same size between the same two price levels. This makes the up-versus-down comparison within-run and matched, which matters: under a random-jump process the per-run noise on this statistic has standard deviation 0.74, large enough that a single seed of the symmetric baseline returned a spurious  $-0.82$  gap.

Each trader’s private value is  $F_t + \varepsilon_{i,t}$  with  $\varepsilon \sim \mathcal{N}(0, 5)$ , redrawn each period. All experiments here use full information; the delayed and noisy signal treatments are implemented but not reported.

## 3.3 The decision contract

Code owns matching, budgets, order sizes and prices. An agent answers three typed questions and nothing else: direction (**buy/sell/pass**), urgency on a five-level scale, and whether the signal is already reflected in the book. A code-owned rule maps urgency to an integer tick, anchored on the book and hard-clamped at the agent’s own private value.

That clamp is load-bearing. *An agent cannot express a losing price*, because it never expresses a price. A 20,000-case property test asserts that no quote is ever on the wrong side of the quoting agent’s value. Consequently every effect we report is about direction and timing, never about arithmetic.

The question wording is frozen and hash-locked; a test fails if any instruction or criterion changes. This is the defence against unrecorded prompt iteration.

## 3.4 Arms

All four arms emit the same object, including a distribution over **{buy, sell, pass}**, which makes them comparable.

- **ZI**: random direction and urgency, fixed policy distribution, value-clamped so it never knowingly trades at a loss.
- **NBR**: logit best-response over the same urgency grid, scored by expected surplus times a fill probability. Exactly symmetric between sides by construction; a buy at mispricing  $+x$  and a sell at  $-x$  receive identical probability.
- **Jev-argmax**: the model’s modal action.
- **Jev-sample**: a draw from the model’s distribution.

The two model arms read one response, so for a given observation they are one API call and differ only in the decode rule.

## 4 The model and its measurement properties

The model is a commercially available typed-decision model that returns, for a multiple-choice question, a selected option together with a probability over all options. Three properties affect measurement and are reported because they constrain what can be claimed.

**Non-determinism.** Identical requests return different answers. Four repetitions of one request returned urgency scores of 2.84, 2.79, 2.82 and 2.74 and buy probabilities of 0.98, 0.99, 0.99, 0.99. There is no seed parameter. Run-to-run noise is roughly  $\pm 0.05$  on a 0–4 urgency scale and  $\pm 0.01$  on probabilities. *Reproducibility therefore requires the response archive, not merely the code:* a rerun without it does not reproduce. All 110,805 responses are retained and every reported figure is regenerable from them.

**Quantisation.** Probabilities are returned to two decimal places, so three-option distributions routinely sum to 0.99 or 1.01. We accept and renormalise within tolerance.

**Undocumented failure modes.** The published status codes are incomplete; the service returns bare 5xx from its edge under load. A client written to the documentation alone will treat these as fatal.

## 5 Experiment 1: wording, conviction, and a null

### 5.1 Domain wording induces a conviction asymmetry

Our first criteria were written in ordinary domain language: *buy* as “the good is underpriced... you want to acquire a unit”, *sell* as “the good is overpriced... you want to dispose of a unit”. At equal absolute mispricing the model placed 0.988 on the profitable side when buying was correct and 0.738 when selling was correct, a gap of 0.250, with the missing mass on *pass*.

The gap survives a second price level (+0.225), reversing the option order (+0.235), and renaming the options neutrally (+0.185). It does not survive rewriting the two criteria as exact structural mirrors—“value is *above* the price” versus “*below* the price”—under neutral instructions, where the gap is +0.005 (Figure 1).

Two points matter for practice. First, the modal action is *correct on both sides at every point tested*: a practitioner reading only the selected option sees nothing wrong. Second, the wording that induces the skew is the natural one; symmetry had to be engineered deliberately.

### 5.2 It does not move pricing error

We predicted this skew would slow price discovery after negative shocks. A four-seed exploratory run supported it (+0.63, four of four seeds positive, Holm  $p = 0.061$ ). We registered the test and ran it on twenty fresh seeds. It failed:  $-0.184$ , 95% CI  $[-0.479, +0.111]$ , seven of nineteen seeds positive, one-sided  $p = 0.897$ . The confirmatory interval excludes the exploratory estimate.

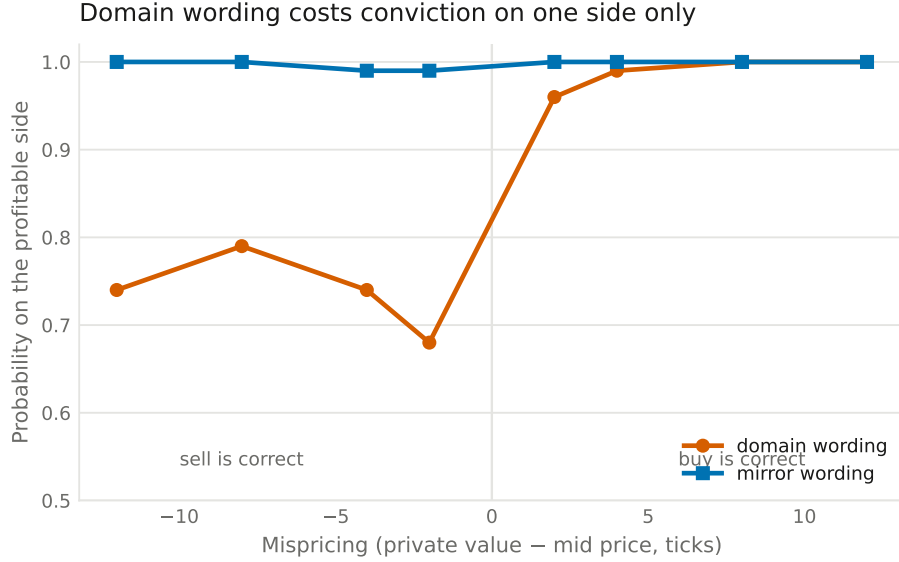


Figure 1: Probability the model places on the profitable side, against mispricing, under the two wordings. The model is symmetric when asked symmetrically. The effect is  $25\times$  the model’s own run-to-run noise.

With  $SE = 0.140$  the run had ample power to detect the exploratory effect, so this is evidence of absence.

We report it as such. Wording skews what the model says; it does not, in this market, change what prices do.

## 6 Experiment 2: liquidity failure under modal decoding

The null above was correct for the question it asked, and the question was wrong. Root mean squared pricing error is defined only over periods in which trade occurred, and so discards exactly the periods where the effect lives. In one run, modal decoding produced zero trades in all three up-jump windows and full trading in all three down-jump windows—270 trades in the run, none in the 45 periods after a positive shock.

We therefore registered a new outcome, defined for every period:  $\text{traded}(d)$ , the share of the fifteen periods after a jump of direction  $d$  in which at least one trade occurred.

**Mechanism, observed.** We instrument resting depth on both sides at the close of every period, which separates two failures: *no counterparty* (one side empty) and *no crossing* (both sides present, prices do not meet). Table 2 classifies every silent post-shock period.

The failure is unambiguous and one-sided: the market runs out of *sellers*. It is never short of buyers in any arm.

**Two components, not one.** Unanimity alone does not account for the direction of the effect, and we report this because our own account of it was initially wrong. Measured directly, the

| Arm        | Traded share |            | Gap   |
|------------|--------------|------------|-------|
|            | after up     | after down |       |
| Jev-argmax | 66.6%        | 93.8%      | +25.8 |
| Jev-sample | 93.0%        | 93.9%      | +3.1  |
| ZI         | 77.3%        | 75.0%      | -0.4  |
| NBR        | 99.8%        | 99.3%      | +1.0  |

Table 1: Preregistered confirmatory test, twenty fresh seeds. D1: the Jev-argmax gap exceeds zero,  $t = 6.37$ , eighteen of twenty seeds positive. D2: it exceeds the Jev-sample gap by 22.7 points,  $t = 6.01$ . Both Holm-significant.

| Arm           | Shock         | No seller | No buyer | Both sides, no cross |           |       |               |       |       |     |  |  |  |
|---------------|---------------|-----------|----------|----------------------|-----------|-------|---------------|-------|-------|-----|--|--|--|
| Jev-argmax up | <b>100.0%</b> | 0.0%      | 0.0%     | down 96.2%           | 0.0%      | 3.8%  | Jev-sample up | 71.4% | 4.3%  |     |  |  |  |
| 24.3%         | down 57.1%    | 7.1%      | 35.7%    | ZI up 0.9%           | 2.8%      | 96.3% | down 2.3%     | 1.8%  | 95.9% | NBR |  |  |  |
|               |               | up 8.3%   | 0.0%     | 91.7%                | down 0.0% | 0.0%  | 100.0%        |       |       |     |  |  |  |

Table 2: Why each silent post-shock period was silent, twenty seeds. Under modal decoding the market is missing a counterparty essentially always; the symmetric baselines fall silent the ordinary way, with both sides quoting and prices failing to meet.

agreement rate is *higher* after down shocks (96%) than up shocks (92%), while the halt is far worse after up shocks. A pure unanimity story predicts the opposite ordering.

Order flow resolves it (Table 3). The modal arm carries a standing buy bias, and the two shocks interact with it differently. After an up shock the bias compounds with the shock and order flow becomes overwhelmingly one-sided. After a down shock—where every trader’s value lies below the prevailing price and selling is plainly indicated—the arm still issues *more buys than sells*. It never reaches a sell majority. The bias and the shock roughly cancel, flow stays mixed, and the market continues to clear.

So the effect decomposes. **Unanimity** sets its magnitude and produces the size-scaling of Section 7: with more traders, the chance that every one of them agrees falls, and the halt fades. **A directional bias** sets which shock it strikes, by determining whether the bias reinforces or offsets the shock. The sampling arm exhibits the same bias in muted form (41.2% versus 39.0% after a down shock), consistent with the falsified E2 of Section 7: sampling dilutes correlation rather than removing it.

**A control that does not quite hold.** NBR returns +1.0 points ( $t = 3.33$ ), small but distinguishable from zero, indicating a slight genuine structural tilt in the market. It is 1/26 of the effect under test, ZI is clean ( $-0.4$ ,  $t = -0.22$ ), and it does not survive the per-size controls of Section 7. We report it because the registration required controls to sit on zero and this one does not.

### 7 Experiment 3: the mechanism, tested where it could fail

If the market halts because every trader reaches the same conclusion, the effect must depend on the number of traders. With agreement rate  $p$  measured independently at 0.92, the probability that all  $N$  agree is approximately  $p^N$ , which falls from 0.72 at  $N = 4$  to 0.07 at  $N = 32$ . We

| Arm        | Shock | Buy   | Sell  | Pass  |       |              |              |       |            |       |       |       |       |  |
|------------|-------|-------|-------|-------|-------|--------------|--------------|-------|------------|-------|-------|-------|-------|--|
| Jev-argmax | up    | 65.9% | 28.2% | 5.8%  | down  | <b>48.8%</b> | <b>42.3%</b> | 8.8%  | Jev-sample | up    | 44.3% |       |       |  |
|            |       | 34.9% | 20.8% |       |       |              |              |       |            |       |       |       |       |  |
|            | down  |       | 41.2% | 39.0% | 19.8% | ZI           | up           | 46.2% | 44.1%      | 9.8%  | down  | 44.8% | 45.9% |  |
|            |       |       |       |       |       |              |              |       |            |       |       |       |       |  |
|            |       | 9.3%  | NBR   | up    | 51.6% | 48.4%        | 0.0%         | down  | 47.9%      | 52.1% | 0.0%  |       |       |  |

Table 3: Order flow in the fifteen periods after a shock. Both algorithmic baselines respond to shock direction symmetrically. The modal arm does not: it issues more buys than sells even after a down shock, which is why the down-side halt is mild and the up-side halt severe.

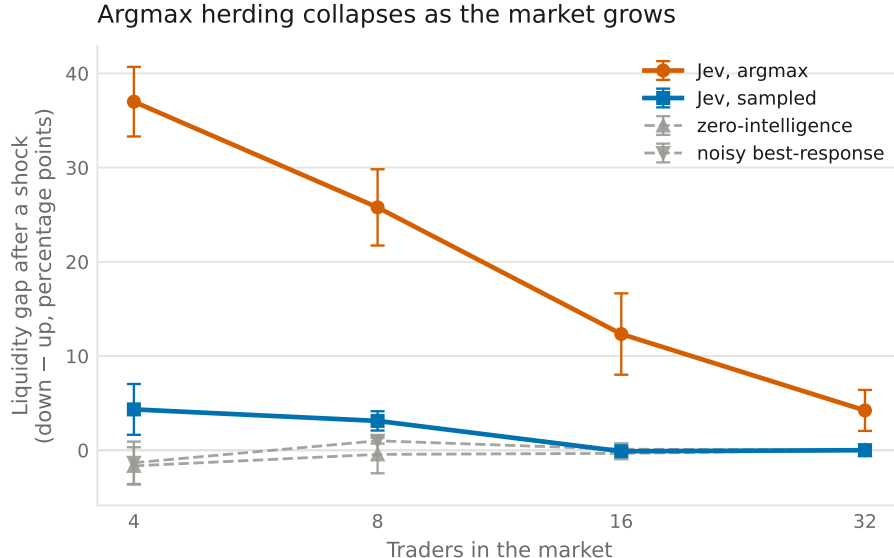


Figure 2: The liquidity gap against market size, twenty seeds per point, bars are standard errors. At  $N = 32$  both symmetric baselines trade in 100% of periods with a zero gap, so there is no structural floor and the residual 4.2 points is attributable to the arm.

registered two predictions before running: that the argmax gap would decay with  $N$  (E1), and that the sampling gap would not (E2).

**E1 confirmed.** The gap falls from 37.0 points at  $N = 4$  to 4.2 at  $N = 32$ ; slope  $-0.112$  per doubling,  $t = -6.91$ . It is robust to dropping the  $N = 4$  point, which we flagged in advance as floor-constrained because the symmetric baselines themselves trade in only 33% of periods there ( $-0.108$ ,  $t = -4.21$ ).

**E2 falsified.** We predicted sampling would show no decay. It shows  $-0.016$  per doubling,  $t = -2.53$ . The available rescue—attributing it to the floor-constrained  $N = 4$  point—fails: dropping that point makes the decay *more* significant ( $t = -3.53$ ).

The corrected account is better than the prediction. Sampling herds as well, seven times more weakly, on the same curve. At a buy probability of 0.99, most sampled traders still choose to buy; sampling does not abolish correlation but dilutes it. There is one mechanism, whose strength is set by how sharply the decode rule collapses the distribution, rather than two regimes.

## 8 Discussion

The practical claim is narrow and, we think, general in shape: *the best way to use one model is not the best way to use many of them*. Modal decoding is individually optimal and collectively degenerate whenever agents share information and require counterparties or diversity. Sampling is individually worse—it deliberately takes a less likely action some of the time—and here it was the difference between a market that cleared and one that did not.

The failure is not a reasoning failure. The model’s modal action was correct on both sides at every mispricing tested. Nothing in its judgement was wrong; the loss occurred in aggregation. Evaluation of a single agent’s decisions cannot detect this, which is what makes it worth reporting.

## 9 Limitations

**The origin of the directional bias.** Section 6 establishes *that* the modal arm carries a standing buy bias, and that this bias determines which shock the halt strikes. It does not establish *why* the bias exists. Both symmetric baselines are balanced at every market size, so it is not an artefact of the market’s construction, but we cannot presently attribute it to the model, to the task framing, or to the state representation.

**One model.** All results are for a single typed-decision model. We have no evidence about others, and cannot say whether the agreement rate that drives the mechanism is characteristic or idiosyncratic.

**Provenance of the probabilities.** We treat the returned probabilities as the model’s beliefs. We have asked the vendor whether they are calibrated posterior mass or post-processed, and have no answer. If the latter, Experiment 1 is a statement about a serving stack rather than a model.

**Full information only.** Every trader observes  $F_t$  exactly, which maximises the correlation the mechanism exploits. The delayed and noisy treatments are implemented but unreported, and would plausibly weaken the effect.

**Scale.** Markets of four to thirty-two traders over 160 periods. The extrapolation to larger markets is the fitted decay, not a measurement.

## 10 Reproducibility

Because the model is non-deterministic and has no seed, code alone does not reproduce these numbers. Every response is archived—110,805 of them—and the analysis can be replayed against the archive with live calls disabled, in which case a cache miss raises rather than silently contacting the API. Total spend was approximately \$3.30 across roughly 95,000 live calls.

The preregistrations, including the falsified hypotheses and the reasons the outcome measure changed between Experiments 1 and 2, are in the repository with their commit timestamps.

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